

PROJECT REPORT

Campus Lab Equipment Manager

1. Introduction

Laboratories contain many types of equipment used by students and faculty members. Managing these items manually can make it difficult to know which equipment is available and which equipment has been issued.

Campus Lab Equipment Manager is a Python-based application developed to provide a simple method of maintaining laboratory equipment records. The application allows users to add, view, search, issue, return, and summarize equipment.

2. Problem Statement

Develop a Python program that can maintain laboratory equipment records and provide basic equipment issue and return functionality.

3. Objectives

- Maintain laboratory equipment records.
- Add new equipment to the system.
- Display all registered equipment.
- Search equipment using its name or category.
- Track whether equipment is available or issued.
- Record the student to whom equipment is issued.
- Process equipment returns.
- Display an inventory summary.

4. Scope of the Project

The project is designed for a small college laboratory. It can be used to maintain basic information about equipment and its current status. The project focuses on fundamental Python programming concepts and does not require an external database or additional software libraries.

5. Technologies Used

Technology	Use
Python 3	Programming language
Python IDLE	Development and execution
IDLE Shell / CLI	User interaction
External libraries	None required

6. Python Concepts Used

Concept	Application
Variables	Store equipment names, categories, IDs, and student names.
Lists	Store multiple equipment records.
Dictionaries	Represent individual equipment records.
Functions	Separate each major operation into a reusable block.
Conditional statements	Make decisions based on menu choices and equipment status.
Loops	Search and display multiple equipment records.
Input/Output	Collect user information and display results.
Exception handling	Handle invalid numeric equipment IDs.

7. Main Features

Add Equipment

The user enters the equipment name and category. The program automatically generates an equipment ID.

View All Equipment

Displays ID, name, category, status, and issued-to information.

Search Equipment

Searches by equipment name or category keyword.

Issue Equipment

Changes an available item's status to Issued and records the student name.

Return Equipment

Changes an issued item's status back to Available.

Equipment Summary

Displays total, available, and issued equipment.

8. Algorithm

1. Start the program.
2. Create an empty equipment list.

3. Display the main menu.
4. Ask the user to select an option.
5. Perform the operation associated with the selected option.
6. Update equipment status when an item is issued or returned.
7. Display the result.
8. Return to the main menu.
9. Stop when the user selects Exit.

9. Sample Test Cases

Test Case	Input	Expected Result
Add equipment	Name + category	Equipment is added
View equipment	Option 2	All equipment is displayed
Search	Name/category	Matching records are displayed
Issue available item	Valid ID	Item becomes issued
Issue issued item	Already issued ID	Warning is displayed
Return item	Valid issued ID	Item becomes available
Invalid menu	Number outside 1-7	Invalid-choice message
Invalid ID	Non-existing ID	ID not found message

10. Advantages

- Simple and easy to use.
- Suitable for beginners.
- Requires no external libraries.
- Demonstrates fundamental Python concepts.
- Can be extended with additional features.

11. Limitations

- Data is stored only while the program is running.
- No login system is included.
- No database is used.
- Issue/return dates are not tracked.
- Designed for a single user.

12. Future Enhancements

- Permanent file or SQLite storage
- Student ID management

- Issue and return dates
- Due-date and overdue tracking
- Administrator login
- Equipment editing and deletion
- CSV inventory reports
- Graphical user interface

13. Conclusion

Campus Lab Equipment Manager demonstrates how basic Python programming can be used to solve a practical problem in a college laboratory. The project combines lists, dictionaries, functions, loops, conditional statements, user input, and error handling to create a functional equipment-management system.

The application is intentionally simple and can serve as a foundation for future improvements such as permanent storage, authentication, reporting, and database support.

14. Execution

Open Python IDLE, open **Campus_Lab_Equipment_Manager.py**, press **F5**, and follow the menu instructions.